# Gayatri Vidya Parishad College of Engineering Autonomous

Mechanical Engineering | Course Code: ME201

## Mechanics of Solids - Semester Exam Question Bank & Worked Solutions

---

Syllabus Focus: ME201 High-Yield Semester Exam Question Bank & Model Solutions

Summary: Comprehensive question bank with previous semester university examination questions, m

### I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

**Unit 1: Stress, Strain & Elastic Constants**

" Syllabus Scope: Hooke law, Poisson ratio, Modulus of elasticity, Bulk modulus, Shear modulus, Thermal s

**Unit 2: Shear Force & Bending Moment Diagrams**

" Syllabus Scope: SFD and BMD for cantilevers, simply supported and overhanging beams under concentrated

# ME201: Mechanics of Solids - Advanced Topics & Assessment

Academic Track: Mechanical Engineering - Units 3, 4, 5 & Examination Matrix

---

## II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

**Unit 3: Flexural & Shear Stresses in Beams**

" Theory of simple bending, Flexure formula sigma/y = M/I = E/R, Section modulus, Shear stress distribution ac

**Unit 4: Torsion of Circular Shafts & Springs**

" Torsion formula T/J = tau/r = G*theta/L, Polar section modulus, Power transmission in solid and hollow shaft

**Unit 5: Principal Stresses & Column Buckling**

" Mohr circle of plane stress, Principal stresses and planes, Maximum shear stress, Euler buckling formula for

## III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Plot SFD and BMD for a simply supported beam with a uniformly distributed load and calculate the point
2. Derive the Euler critical buckling load P_cr = pi^2 * E * I / L_eff^2 for a column with one end fixed
3. Draw Mohr Circle for an element subjected to sigma_x = 80 MPa, sigma_y = -40 MPa, and tau_xy = 30 MPa,

## IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* S. Timoshenko - Strength of Materials CBS Publishers
* Ferdinand P. Beer, E. Russell Johnston - Mechanics of Materials McGraw-Hill 7th Ed

*Authorized NoteVault Educational Document " Verified for Registered University Students*